

IoT Based: Air Quality Index and Traffic Volume Correlation

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Abstract—Major problem facing urban areas today is air pollution. Gas emissions from cars are considered the most important source of this kind of pollution. Pollutant gases emitted as parts of car exhaust consist of chemicals such as carbon monoxide (CO), nitrogen dioxide (NO₂), ozone (O₃), particulate matter (PM), and Sulphur dioxide (SO₂). The environmental Protection Agency (EPA) guides to measure these chemicals by several methods to calculate the gases' concentration. An Internet of Things (IoT) device is used to monitor air quality in real-time is also described in this paper. It uses a set of sensors that measure air quality at the street level. This paper determined the relationship between traffic volume and the Air Quality Index (AQI) as defined by EPA guidelines. Multiple Linear Regression (MLR) is used to create a mathematical model for the relationship between traffic volume and the Air Quality Index (AQI). This model has been tested on one of the streets in the city of Melbourne, Florida.

Keywords—Internet of Things (IoT), Air Quality Index (AQI), Multiple Linear Regression (MLR), Air Pollutants, and Traffic Volume.

I. INTRODUCTION

Air pollution has become a major problem in different countries due to industrialization in production. This has led to an increase in the number of health problems and premature deaths. Air pollution occurs when smoke or other harmful gases, mainly carbon, nitrogen, or sulfur oxides, become suspended in the air [1].

Significant air pollution sources are combustion from industries, emissions from transport, agriculture side effects, tobacco smoke, forest fires, and home heating. In addition to gaseous pollutants, air pollution may contain solid particles. These solid particles may come from car emissions, chemicals from factories, and dust. Ozone is also a major part of air pollution, especially in big cities [2]. Air pollution harms on human health, and it may severely damage the ecosystem. It may cause diseases, allergies, and even death to humans; it may harm other living organisms such as animals and food crops and damage the natural or built environment. Both human activity and natural processes can generate air pollution [1, 3].

To show the extent of pollution, pollution and the time of pollution need to monitor air pollution. IoT technology has introduced a series of relatively inexpensive air quality and pollution sensors to monitoring air quality.

This work is aims to quantify the relationship between traffic volume and the Air Quality Index (AQI). A mathematical model for the relationship is developed and using measurements collected by a set of sensors. A block diagram of the sensor used in the study is shown in Fig.1. It consists of sensors that monitor five gases (O₃, SO₂, NO₂, and CO), particle matter (PM), and environmental parameters.

A. Monitoring Air Pollution

The need for air quality monitoring is paramount as air pollution has adverse effects on human health in both short and long terms. Monitoring the air is to find out the degree of air pollution. Long-term data can uncover patterns that can help support air pollution prevention policies [4, 5, and 6].

Air pollution monitoring itself does not reduce air pollution, but it does show the extent of pollution, the location of pollution, and the time of pollution.

Fixed air quality monitoring equipment is used in cities and towns to measure pollution levels at specific locations [1, 6, 7].

There is no single definition of what the Internet of Things (IoT) truly is; in any case, we can characterize or define the IoT by expounding what the IoT can give. The IoT is believed to be the following development of the Internet. It is going to provide a networking infrastructure allowing trillions of devices to collect data and communicate together. [1, 8]. In common definition, the (IoT) is a system of physical objects such as sensors and connectivity that enables transfer, collection, and data exchange. Nowadays, the IoT is regarded as one of the most significant technologies [9, 10].

Use of the IoT technology for mobile air quality monitoring reduces costs, extends coverage, and provides a much higher level of flexibility than traditional fixed solutions. Moreover, IoT ease of deployment can offer several important advantages, such as early indicators of high pollution zones. The development of IoT technology has introduced a series of relatively inexpensive air quality and pollution sensors.

Many low-end IoT sensors lack the high degree of accuracy that stationary stations provide, but they still provide good of air quality and pollution indicator. IoT sensors can naturally send information back to the cloud utilizing some form of cellular communication technologies. The most wide technologies are 3G/4G in some regions; the sensors might have the option to utilize LoRa – a long-range wireless protocol [12]. The main appeal of the IoT is that it can give a complete way to understand the severity of air pollution, better understand the issue and determine measures to set up stronger links between regulators, city counselors, and citizens. Investing in IoT solutions to supervise air quality also has financial benefits for government and city managers. For example, a cleaner air means a lower budget spent on health issues related to air pollution [11, 12].

B. Air Quality Index

The air quality index is used by various governments to inform the public about how clean or polluted the air is. One of the most frequently used air quality indices is the Air Quality Indicator (AQI) developed by US EPA. As shown in figure 1, the AQI gives results through a color-coded scheme and numbers where the colors are linked to the air pollution impact's severity. The quality air index has standards for informing the effect of air quality on human, which principal value lies between 0- 500 [1, 4].

A higher value of the AQI is more dangerous to air quality and vice versa. Some countries have different indices of Air Quality Index (AQI). However, the relevance of the AQI to the public attained through the national uniformity in a country, as there is no international body assigned to these standards. Every government works independently. Nowadays, people are suffering from new diseases due to their choice of working and living conditions.

However, the preventive measure they take to reduce the situations has side effects on them. In the US, using EPA equation to calculate the AQI [13].

$$I_p = \left[\left\{ \frac{(I_{high} - I_{LOW})}{(C_{high} - C_{LOW})} \right\} * (C_p - C_{LOW}) \right] + I_{LOW} \quad (1)$$

Where C_{high} = Breakpoint concentration greater or equal to given Concentration, C_{LOW} = Breakpoint concentration smaller or equal to given Concentration, I_{high} = AQI value corresponding to C_{high} , I_{LOW} = AQI value correspond to C_{LOW} , C_p = the concentration of pollutant p, AQI = Maximum of I_p [1, 13].

World Health Organization (WHO) proposes a coloring scheme that could be universally adopted to better understand the different ranges of air quality. The colors significantly show the index range as GREEN for air pollution is clear and no risk. YELLOW for moderate as one should be acceptable and little risk. ORANGE color shows the unhealthy for a sensitive groups. RED for all unhealthy was decisive and immediate action taken. WHO considers PURPLE for very unhealthy, black; the most hazardous, and extremely high risk for all groups [4].

AQI Category	AQI Value
Good	0 - 50
Moderate	51 - 100
Unhealthy for Sensitive	101 - 150
Unhealthy	151 - 200
Very Unhealthy	201 - 300
Hazardous	>300

Fig 1. AQI ranges [4].

II. LITERATURE REVIEW

Measurement the air quality considers the first step is to solve the pollution problems. This section presents some of the research on using the IoT to measuring the air quality. In [14], the researchers developed an IoT system that can determine the location, detects pollutants in the environment, and collects data in an automated manner. This data are presented as a pollution route map where travelers can view routes with fewer pollutants, which have contributed to various environmental parameters. They distributed more than a thousand wireless sensor nodes along street lights. The sensors are sensing humidity, benzene, light intensity, temperature. They cover the area of the university and its surroundings. They used sensors like MQ4, MQ2, MQ135, PM, Sound, Humidity, Temperature, sharp dust. Also, we used battery, wires for connecting, and a connector. Data are transmitted and stored in a database server from where they may be visualized. The experiment is made for 30 days to route from Dadri to LalKuan to Vihar and back to through Noida via Great Noida. This research provides an IoT system that enhances collection, monitoring, and analyzing air quality data. The user can access the data air quality from different vehicles within an area and provide data showing air quality per road. People are provided with valuable information on air pollutants and location while commuting by using setup that forecasts pollutants using real-time and historical data.

In [15], the authors apply on the air pollution index monitor based on the IoT. IoT devices collect information from different detecting devices and form a network. According to research at Limerick University, an API monitoring system comprises antenna node, figures transmission system, cloud databank, and personal computer client application. The research tested carried out at Limerick University as a regional air monitoring area shows the monitoring system's effectiveness in air pollution index. Data obtained from the antenna node used the PEB and the gas sensor module. Using Sigfox via Sipy the API data transmitted. The program of the node mainly comprises of Sigfox Socket link purposes that recognize interpretation and transfer API data.

The data then sent to Sigfox cloud, where one can view the back-end, and store facts to the database. The test results of air pollution index monitoring research in Limerick University successfully obtained the index of air pollution values, read data, and transferred them to the cloud databank. It also completed the real-time data display on the air pollution index, knot status monitoring, and Historic facts display, among other roles. During the one month of testing the node did not run out of power to advantages of using Sigfox [15].

Therefore, in areas with power outage low supply, the low power area wide network is recommended and more convenient to use, as it does not consume a lot of energy and has lower costs. The API, that built in this research shows that it can efficiently monitor air pollution in an area, has long life service, and maintained easily.

In [16], the research applies the (IoT) to develop a system that can track air pollution. It utilizes a three-layer architecture of IoT, consisting of an application, sensor, and networking layer. It uses five tools, which are the Arduino UNO, which serves as the microcontroller, the ESP8266, which is the Wi-Fi module, Ubidots as the cloud service, gas sensors, and Android system. The integration of all these five tools will ensure that the air monitoring application is effectively collects, storing, and recovering data. It also uses the global positioning system to collect data on the pollution levels in a specific area, and alert individuals approaching these hotspots.

The system’s primary features are a real-time index of the quality of air for a specific place, the timing of various activities for the various recommendations, any dips in air quality, reports for air quality, and map generations. The researchers use three phases to develop the application: detection of the concentration of air pollution, producing an Android app with a user-friendly interface, and prediction of the air quality.

The ability to show data by the IoT detection system on the application involved drawing a route from a user’s source to destination, and then send an appropriate warning if the pollution is high. The researchers identify that the IoT application faces the limitation of carrying out complex computations, and in such a situation, the solution is to use fog rather than cloud computing. The IoT Mobair Android application is particularly useful to motorists who can monitor air quality over an entire route.

III. EXPERIMENTAL SET UP

The work aims to use Multiple Linear Regression (MLR) to create a mathematical model for the street level pollution. The measurement system is placed in one of main streets in the city of Melbourne, Florida, the US as shown in figure 3. This Street is one of the busiest streets in Melbourne. In the first phase of this study, a large data set of pollutant concentration measurements is collected from the air quality sensor located in the proximity road.

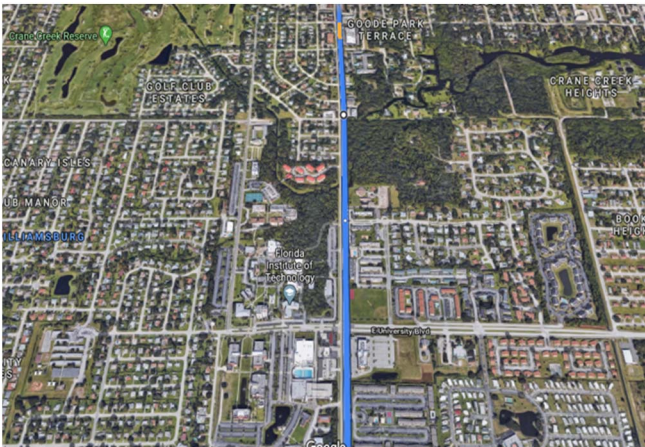


Fig 2. The main street in Melbourne city.

A. Hardware

Utilizing an Internet of Things IoT device, represents Air Quality Sensor that assists in monitoring the air quality in real time. The Air Quality Sensor system which used consists of sensors that provide several environmental parameters monitoring and concentration of the pollutants. The environmental parameters such as air temperature, humidity, and pressure. Additionally, there are sensors for measurements of gas concentrations. Then collects data and sends them via Ethernet, Wi-Fi, cellular (2G/3G/4G) or NB-IoT interface to save or process the data (see figure 3). Also, using Hikvision camera, and NVR to record videos helps collect the number of vehicles in the street during the 10 days. The camera NVR has Hard disk for recording a video for a long time, and able to back to video at any time.

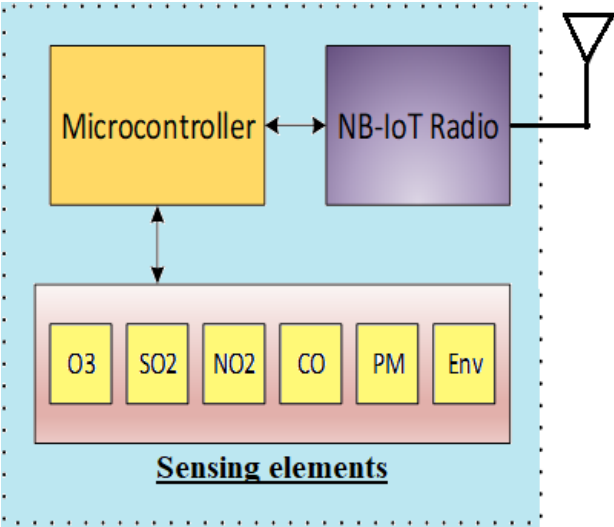


Fig 3. Block diagram of AQ sensor.

In the second phase, the air pollution data are analyzed for each of the pollutants. By comparing the average of the number of cars for an hour with the average of the air quality data for an hour, correlations between them are calculated.

A mathematical model is developed that shows how the traffic density, humidity, pressure, and temperature impact air quality. The mathematical model is based on Linear regression analysis. It predicts the value of a dependent variable as a function of the set of independent variables. The independent variables in this scenario are Traffic Density (TD), Temperature (T), Humidity (H), and Pressure (P). The mathematical model for Multivariate Linear Regression (MLR) is given as:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \epsilon \quad (2)$$

Where, Y is the air quality index for all the air pollution concentrations (AQI, O3, CO, SO2, NO2, PM2.5, and PM10). $\beta_0, \beta_1, \beta_2, \beta_3,$ and β_4 are regression coefficients which unknown, and need to be determined from measured data and values. $X_1, X_2, X_3,$ and X_4 are the independent variables Traffic Density (TD), Temperature (T), Humidity (H), and Pressure (P). ϵ is the error between measurements, and model predictions [12]. Collecting the data which consists of all air pollutant values. This data are provided AQI as an average of concentration for each hour of all pollutants. The AQI values for each of the pollutants are calculated using the equation provided by EPA guidelines, as mentioned earlier. The AQI value selected from the maximum value of all air pollution O3, CO, SO2, NO2, PM2.5, and PM10 [17]:

$$\text{AQI max} (\text{O3AQI}, \text{COAQI}, \text{NO2AQI}, \text{SO2AQI}, \text{PM2.5AQI}, \text{PM10AQI}) \quad (4)$$

B. Experimental Results and Analysis

Measurements of the Air Quality Index for pollutants were collected for 9 hours per day from 8 am to 5 pm over a 10-day period. The mathematical model to calculate and measure the Air Quality Index of air pollution. We used a multiple linear regression (MLR) in our experiment to analyze our data, because we have more than two variables; air pollution, and Traffic Density, Temperature, Humidity, and Pressure. After we collect air pollution data and the number of cars in the street, we then create the multiple linear regression model as shown in table 1 for all air pollution. Where : TD: Traffic Density, T: Temperature, H: Humidity, P: pressure. Tables 1 and 2 present the Multiple Linear Regression (MLR) model for all air pollutions.

TABLE 1. Multiple Linear Regression Model

Air Pollution	Multiple Linear Regression Model
AQI	$\text{AQI} = 62.72 + (-0.014 * T) + (-0.076 * H) + (-0.117 * P) + (0.041 * TD)$
O3	$\text{O3} = -717.8 + (0.45 * T) + (-0.179 * H) + (0.667 * P) + (0.033 * TD)$
CO	$\text{CO} = -13.4 + (0.008 * T) + (0.027 * H) + (-0.004 * P) + (0.008 * TD)$
SO2	$\text{SO2} = -598 + (0.48 * T) + (0.024 * H) + (0.54 * P) + (0.021 * TD)$
NO2	$\text{NO2} = -272 + (-0.123 * T) + (0.2166 * H) + (0.273 * P) + (-0.0026 * TD)$
PM2.5	$\text{PM2.5} = 4115 + (-1.29 * T) + (0.50 * H) + (-4.13 * P) + (0.026 * TD)$
PM10	$\text{PM10} = 1090 + (-0.28 * T) + (0.148 * H) + (-1.10 * P) + (0.0078 * TD)$

TABLE 2. Result of MLR Model

Regression Statistics				
MLR Model	Multiple R	R Square	Adjusted R Square	Standard Error
AQI	0.87	0.76	0.75	2.51
O3	0.82	0.67	0.66	3.51
CO	0.77	0.59	0.57	0.64
SO2	0.65	0.42	0.40	2.90
NO2	0.83	0.69	0.67	3.06
PM2.5	0.62	0.39	0.36	6.83
PM10	0.67	0.45	0.42	1.79

Using this model, we can see a strong correlation between independent variables such as temperature, humidity, pressure, and traffic density plus air pollutions. The MLR models documented in Table 1 show that the traffic volume impacts AQI, NO2, O3, and CO the most.

Besides, Table 3 below shows the R Square results of Simple Linear Regression (SLR) and Multiple Linear Regression (MLR) for all air pollutions. The R square value in Simple Linear Regression (SLR) tells us that there is a correlation between traffic density and air pollutions. The R square value in Multiple Linear Regression (MLR) tells us that there is a correlation between the dependent variable and independent variable of the air pollutions. As seen in Table 3 and figure 4, the results of Simple linear regression showed that the number of cars is very little effected the NO2, SO2, PM2.5, and PM10. The R Square in MLR Result showed that the NO2, PM2.5, and PM10 were remarkably affected by the independent variable such as temperature, humidity, pressure, and Traffic density.

TABLE 3. R Square of SLR Model and MLR Model

SLR Model	R Square	MLR Model	R Square
AQI	0.71	AQI	0.76
O3	0.60	O3	0.67
CO	0.50	CO	0.59
SO2	0.35	SO2	0.42
NO2	0.06	NO2	0.69
PM2.5	0.02	PM2.5	0.39
PM10	0.03	PM10	0.45

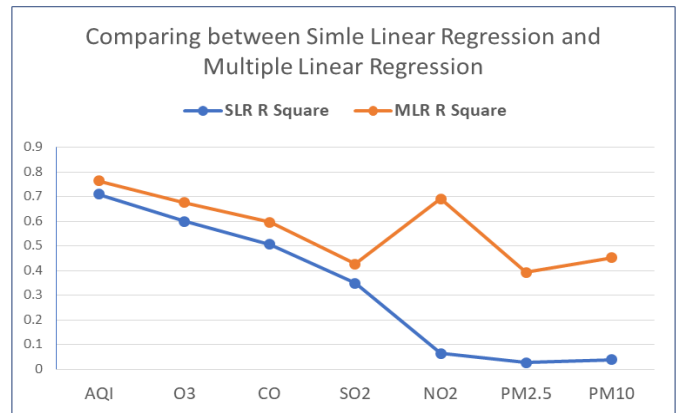


Fig 4. Comparing between SLR and MLR.

IV. CONCLUTIONS AND FUTURE WORK

In this paper, the results presented the collected data that shows how gas emissions from car exhaust and other factors in open areas, such as humidity, temperature, and pressure, will affect air quality. This is achieved by analyzing the air pollution data using multiple linear regression on data collected in an open area. In future work, we will collect other data in different environments, including indoor areas such as parking, and tunnels.

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